

Collaborative Co-located Mixed Reality in Teaching Veterinary Radiation Safety Rules: a Preliminary Evaluation

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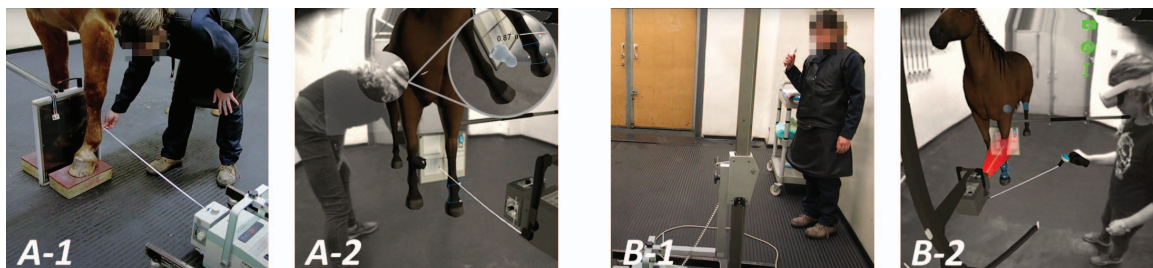


Figure 1: Measuring distance between x-ray tube and image detector (Fig. 1-A-1). Measuring distance in MR (Fig. 1-A-2). Taking radiographs from a safe distance (Fig. 1-B-1). Taking radiograph in MR (Fig. 1-B-2).

ABSTRACT

When learning radiographic techniques on horses, veterinary students must be familiar with radiation safety rules, which are essential to avoid potentially dangerous exposure to the x-ray beam. We propose a collaborative co-located Mixed Reality (MR) system for training students on radiation safety rules allowing staff to guide the students to practice positioning for radiography safely. The performance of 22 veterinary students was measured before and after they had grouped training sessions using the system. The result showed that the participants performed significantly better in the post-test. This evaluation illustrates the potential of expanding the technique into other subjects.

Index Terms: Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human computer interaction (HCI)—Interaction techniques—Gestural input

1 INTRODUCTION

Diagnostic imaging techniques are an essential component of the clinical workup of pathological conditions in both humans and animals. During their medical and veterinary degrees, students are trained in particular on the safe use of radiography machines to obtain the most common radiographic projections. However, while medical students can practice the main radiographic projections through role-playing as patients (in Human Medicine Education [3]), in the veterinary world, practical sessions might be affected by the availability of animals (in particular horses and cows). The need to sedate horses to perform radiographic procedures poses ethical dilemmas regarding using animals for repeated exposures purely for educational purposes. Furthermore, having the time and resources to teach radiography in university teaching hospitals with a busy caseload is extremely difficult. To ensure radiation safety protocols, a limited number of students are allowed in the room when performing routine radiographic studies on large animals. Therefore, although it is important for veterinary students to learn the basics

of radiographic positioning and image acquisition, this can be challenging in university settings. Any technology that can reduce this resource burden should be pursued vigorously.

Virtual simulation has drawn researchers' attention as a potential training method in medical imaging, thereby producing a safer environment. More recently, Head-mounted-display (HMD) based Virtual Reality (VR) and Mixed Reality (MR) have been used to overcome the limitations encountered in many aspects and subjects, including medical education [8]. Compared with conventional 2D methods, the HMD with hand tracking provides an immersive experience and engaging interactions to simulate reality [9]. There have been applications of immersive VR in training human radiographic techniques [2, 4–6], although none of those studies intentionally addressed the importance of the radiation safety rule during the intervention. In the previous study, students felt VR did not contribute to their knowledge of radiation safety as no related information was displayed or taught to them [3].

A hand-tracking passthrough MR system for teaching radiation safety rules during the process of demonstrating radiographic techniques in horses is proposed [7]. We described the features designed specifically for safety training. Students can use this system to practice safely using the equipment without exposure to ionising radiation for themselves, other operators and the animals. Meanwhile, they will gain an understanding of why it is dangerous for everyone in the room when the safety rule is violated through the visualisation of the radiation. The system will also allow them to revise at their own pace and location with real-time guidance from qualified staff.

We conducted a user study to evaluate the proposed system and discussed whether it would contribute to students' knowledge about radiation safety rules. The students were allocated into small groups and had a face-to-face safety teaching session delivered by the radiology course lecturer. The score of a customised MCQ was measured before and after the participants had group training sessions.

2 IMPLEMENTATION

A minimum of three operators in the veterinary radiology institution are needed when dealing with large animal radiography. The student needs to be concerned not only about themselves but also about other people on-site. So the teaching system has to support a multi-user function to achieve the best teaching quality.

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General radiation safety rules are taught in the first week of the clinical rotation as well as in other lectures throughout the veterinary medical degree. However, there is no individual training session designed only for safety training. The system was designed to train students on basic equine radiographic techniques and to demonstrate the value of implementing radiation safety rules while highlighting the potential risks involved in the procedure. All the users can directly manipulate virtual objects using hand gestures (pinching or grabbing). In the beginning, the lecturer will verbally deliver basic safety information, such as the minimum age to be allowed in the equine radiography room, and the number of people allowed in the room when obtaining radiographs of a horse.

After positioning the image detector at the correct anatomical position, the distance between it and the x-ray tube is measured using the ruler on the machine (Fig. 1-A-1). The same function has been implemented in the simulation (Fig. 1-A-2). When a user grabs the ruler, the distance figure that is the same as the reality will show up on the top of the ruler handle. The second step is to ensure that the beam is correctly collimated. Finally, the user takes a radiograph by using an extendable exposure button attached to the x-ray machine (Fig. 1-B-1) and stands as far as possible from the x-ray machine and the horse. The same procedure is implemented in the MR space, where the exposure button can be grabbed with one hand and pressed with the index finger of the other hand (Fig. 1-B-2).

When the exposure button is pressed and all the conditions meet the requirements, the system will display the corresponding radiograph instantly on a canvas. This function provides instantaneous feedback to the students, letting them know whether they are correctly following the radiographic procedure or not. The hint mode can visualise all joints' positions and a ray representing the central beam. It can be challenging for a student with little to no experience to correctly centre the beam to the area of interest since soft tissues and fur cover joints in reality. The hint mode provides an easier way to correctly set up the x-ray machine, facilitating student training and shortening teaching effort and time. Once the students are familiar with the radiographic procedure in hint mode, the instructor can turn it off to allow for a more realistic practice experience.

3 RESULTS

Twenty-four veterinary students who were enrolled in the weekly clinical rotation responded to participate, which was 20% of the total students enrolled (86 UGY and 34 GY) in the programs. Two of them withdrew: one due to a personal situation; the other had previous cybersickness symptoms using VR HMD. In the end, 22 veterinary students (male:female, 7:15) voluntarily participated in the study. Most of them had limited experience in VR gaming, and the rest occasionally played VR games. Due to the demography of the veterinary medicine school, some of the students had previous experience with horses and some had none. They were either enrolled in undergraduate or graduate programs, which made no difference to their course or the knowledge background requirements. There were 11 safety training sessions successfully conducted. Nine of those had two students as a group, while one session had one student due to a participant dropping out, and one session had three students due to a volunteering on-site.

The Shapiro-Wilk test showed a non-normal distribution for the pre-test and post-test scores ($p = 0.020$). The distribution of the post-test score was a skew normal distribution. The resulting scores are shown in Fig. 2. Wilcoxon signed-rank test was performed for pairwise significance testing. There was a significant difference ($***$) ($p < 0.001$) between pre-test ($\bar{x} = 71.818\%$; $\sigma = 15.625\%$) and post-test ($\bar{x} = 85\%$; $\sigma = 10.579\%$).

4 DISCUSSION

This study suggests that it is possible to use a co-located MR system to teach multiple students about radiation safety regulations and ra-

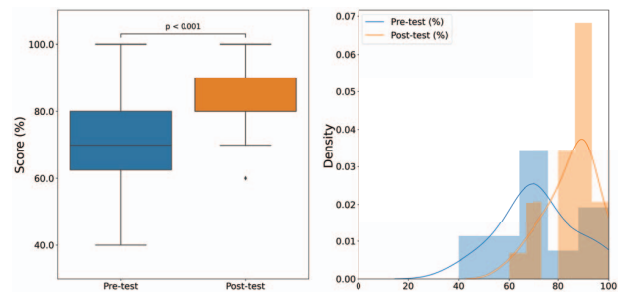


Figure 2: Box plot and distribution plot of radiation safety rule test score before and after the training session.

diographic techniques at the same time. By utilizing the pass-through cameras on the VR HMD, students can see their surroundings and interact with other users on the site, enhancing communication and improving the efficiency of the teaching process. This also reduces the load on the graphics rendering, which is beneficial for mobile HMDs [1]. The instructor can teach theory and demonstrate procedures based on their interaction with the students. It is believed that complex scenarios and procedures can be simulated, allowing the instructor to easily administer formative assessments during the training session to gauge the students' retention of the information.

The radiation safety test results revealed that students performed significantly better in the post-test after the MR session. All of them achieved better or equal scores compared to the pre-test result, confirming co-located MR training's feasibility and acceptability.

Future work will be comparing the result of this study including user experience questionnaires, user movement tracking data, and user sickness questionnaires with a simulated teaching system using an actual x-ray machine and horse leg skeleton model.

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